Rank–Nullity

E.g. $T: \mathbb{R}^n \to \mathbb{R}^m$

$$T(\underline{x}) = A\underline{x}$$

$$\underbrace{(\#\text{ of pivot col. of } A)}_{=\operatorname{rank}(T)} + \underbrace{(\#\text{ of free col. of } A)}_{=\operatorname{nullity}(T)} = \underbrace{n}_{=\dim(\mathbb{R}^n)}$$

Proof: $T: V \to W$, $\ker(T) \leq V$

By Complementation Theorem, $\ker(T) \oplus V_1 = V$ $(V_1 \leq V)$ (=> Step ②)

Consider $T|_{V_1}: V_1 \to T(V_1)$ $(= \operatorname{Im}(T|_{V_1}))$

($\because T|_{V_1}(\underline{v_1}) = T(\underline{v_1})$,

i.e., $T|_{V_1}$ is the restrict l. trans. from V_1 to $T(V_1)$ $(= \operatorname{Im}(T|_{V_1}))$

(WTS: $\dim(V_1) = \dim(\operatorname{Im}(T))$

(Step) ① (WTS: $\dim(V_1) = \dim(T(V_1))$ (=> Step ①)

(<= WTS: $T|_{V_1}$ is an iso. of vector spaces)

(1) $T|_{V_1}$ is surj. $\Leftarrow$ $T|_{V_1}: V_1 \to T(V_1)$ (①如果此处为W, 则不一定)

(2) $T|_{V_1}$ is inj.:

Consider $\underline{v} \in \ker(T|_{V_1})$ $\Rightarrow T|_{V_1}(\underline{v}) = T(\underline{v}) = \underline{0}_W$, $\underline{v} \in V_1$

$\Rightarrow \underline{v} \in V_1 \cap \ker(T) \overset{②}{\Rightarrow} \underline{v} = \underline{0}_V$?

$\Rightarrow \ker(T|_{V_1}) = \{\underline{0}\}$

② (WTS: $\dim(T(\underline{v_1})) = \dim(\mathrm{Im}(T)) = \dim(T(V))$ ($\Leftarrow T: V \to W$)

(<= WTS: $\dim(\mathrm{Im}(V_1)) \leq \dim(\mathrm{Im}(V))$)

(<= WTS: $V_1 \leq V$)

(<= WTS: ~~Im~~$T(V_1) \geq$ ~~Im~~$T(V)$)

① ②

Take $\forall\ T(\underline{v}) = T(\underline{v_k} + \underline{v_1}') \in \mathrm{Im}(V)$

① ② $\ker(T)$

($\ker(T) \oplus V_1 = V$)

$\Rightarrow T(\underline{v}) = T(\underline{v_k} + \underline{v_1}')$

(l. trans. ⊗)

$= \underline{0} + T(\underline{v_1}')$

$\Rightarrow T(\underline{v}) \in T(V_1)$

?

① $\Rightarrow T(V) \leq T(V_1)$

① ~~①~~

$\Rightarrow \dim T(V_1) = \dim T(V) = \dim(\mathrm{Im}(T))$

~~①~~ ②

Coordinate Vectors & Matrix Representations

(Goal:
Express a vector $\underline{v} \in V$
by a column vector $[\underline{v}]_{\mathbb{R}} \in \mathbb{R}^n$)

Coordinate Vectors

Def.: V: a finite dimensional v. space
$B = \{\underline{v}_1, \cdots, \underline{v}_n\}$: an ordered basis of V

E.g.: $V = \mathbb{R}^2$, $B = \{\begin{pmatrix}1\\0\end{pmatrix}, \begin{pmatrix}0\\1\end{pmatrix}\}$, $\underline{v} = \begin{pmatrix}2\\1\end{pmatrix} \in V$

$\underline{v} = 2\begin{pmatrix}1\\0\end{pmatrix} + 1\begin{pmatrix}0\\1\end{pmatrix} \Rightarrow [\underline{v}]_B = \begin{pmatrix}2\\1\end{pmatrix}$

$B' = \{\begin{pmatrix}1\\1\end{pmatrix}, \begin{pmatrix}1\\-1\end{pmatrix}\}$

$\underline{v} = \frac{3}{2}\begin{pmatrix}1\\1\end{pmatrix} + \frac{1}{2}\begin{pmatrix}1\\-1\end{pmatrix} \Rightarrow [\underline{v}]_{B'} = \begin{pmatrix}\frac{3}{2}\\ \frac{1}{2}\end{pmatrix}$

Remark:
Fix any $\underline{v} \in V$ and ordered basis B,
the coord. v_i $[\underline{v}]_B$ is uniquely determined

Proof: $\underline{v} = \alpha_1 \underline{v}_1 + \cdots + \alpha_n \underline{v}_n = \beta_1 \underline{v}_1 + \cdots + \beta_n \underline{v}_n \Rightarrow \begin{pmatrix}\alpha_1\\ \vdots \\ \alpha_n\end{pmatrix} = \begin{pmatrix}\beta_1\\ \vdots \\ \beta_n\end{pmatrix}$

Theorem

Let V ~~—~~ and B ~~—~~

Then the func. $[\ ]_B : V \to F^n$, $\underline{v} \mapsto [\underline{v}]_B$

① is well-defined

and is ② an isom.

(In set theory,
a func. $f: P \to Q$ is well-defined
$\Leftrightarrow$ $f(p)$ takes only one value
(and) $f(p) \in Q$
[e.g. ~~$\sqrt{\ }$~~ : $\mathbb{R} \to \mathbb{R}$, $4 \to 2, -2$ — is not well-defined)

Proof ①: ($\Leftarrow$ Remark & ↓)

Proof ②:

(1) (WTS: $[\ ]_B : V \to F^n$ is a (l. trans.
$\Leftrightarrow$ $[\underline{v}+\underline{w}]_B = [\underline{v}]_B + [\underline{w}]_B$)

(2) (WTS: $[\ ]_B$ is surj.)
$\forall \begin{pmatrix} \alpha_1 \\ \vdots \\ \alpha_n \end{pmatrix} \in F^n \Rightarrow [\underline{v}']_B = \begin{pmatrix} \gamma_1 \\ \vdots \\ \gamma_n \end{pmatrix}$
$\Rightarrow \underline{v}' := \gamma_1 \underline{v}_1 + \cdots + \gamma_n \underline{v}_n$

(3) (WTS: $[\ ]_B$ is inj.
$\Leftrightarrow$ ($[\underline{v}]_B = [\underline{v}']_B \Rightarrow \underline{v} = \underline{v}'$))

Ex.: (记号表)

Given 2 ordered basis B, B' of V and $B \neq B'$,

then $\forall \underline{v} \in V$, $[\underline{v}]_B \neq [\underline{v}]_{B'}$

Change of Basis Matrix

Def.: $B = \{\underline{v_1}, \cdots, \underline{v_n}\}$

and $B' = \{\underline{w_1}, \cdots, \underline{w_n}\}$

are 2 ordered basis of V

$\underline{v_i} \in B$, $\underline{w_j} \in B'$

$$\begin{aligned} \underline{v_1} &= \alpha_{11}\underline{w_1} + \cdots + \alpha_{n1}\underline{w_n} \\ &\vdots \\ \underline{v_n} &= \alpha_{1n}\underline{w_1} + \cdots + \alpha_{nn}\underline{w_n} \end{aligned}$$

Change of basis matrix: $P_{B'B} = (\alpha_{ij})$

$$\left(\text{i.e. } \underline{v_j} = \sum_{i=1}^{n} \alpha_{ij}\underline{w_i}\right)$$

E.g.: $V = P_2(\mathbb{R})$, $B = \{1+x+x^2, 2x, -1+3x^2\}$

$B' = \{1, x, x^2\}$

Then $\begin{cases} \cdots = \cdots \\ \vdots \quad \vdots \quad \vdots \end{cases}$,

$P_{B'B} = \begin{pmatrix} 1 & 0 & -1 \\ 1 & 2 & 0 \\ 1 & 0 & 3 \end{pmatrix}$ (注意方向、顺序!)

Prop.: The coord. v. $[\underline{v}]_{B'}$ and $[\underline{v}]_B$ corr. to $\sim$ B', B are related by $\ell_{B'B}[\underline{v}]_B = [\underline{v}]_{B'}$

Corollary: $\ell_{B'B}$ is an invertible $\underbrace{n \times n}_{n = \dim(V)}$ matrix and $(\ell_{B'B})^{-1} = (\ell_{BB'})$

Proof: (只要)